4.5.3 Velocity Data Extraction

Rather than being satisfied with only the full-field salt data and giving a best-guess at the performance of the simulated water cases, the extraction of three distinct velocity profile heights of $Y/H = 0.25$, 0.50, and 0.75 in Figure 14 of Reis et al. [9] was inspired by the experience of performing image pixel-measurements for the modeling effort described in Chapter 5.

Source Figures and Axis Calibration

First, Figure 12 and 14 from Reis et al. [9] were zoomed in on as much as possible to fully capture one of the four subfigures for each figure in-frame. From there, the "source" image is turned into a "calibration" image to ensure that the axes and origin for the figure are set correctly along side the alignment of vertical and horizontal gridlines with the major tickmarks along the x and y axes.

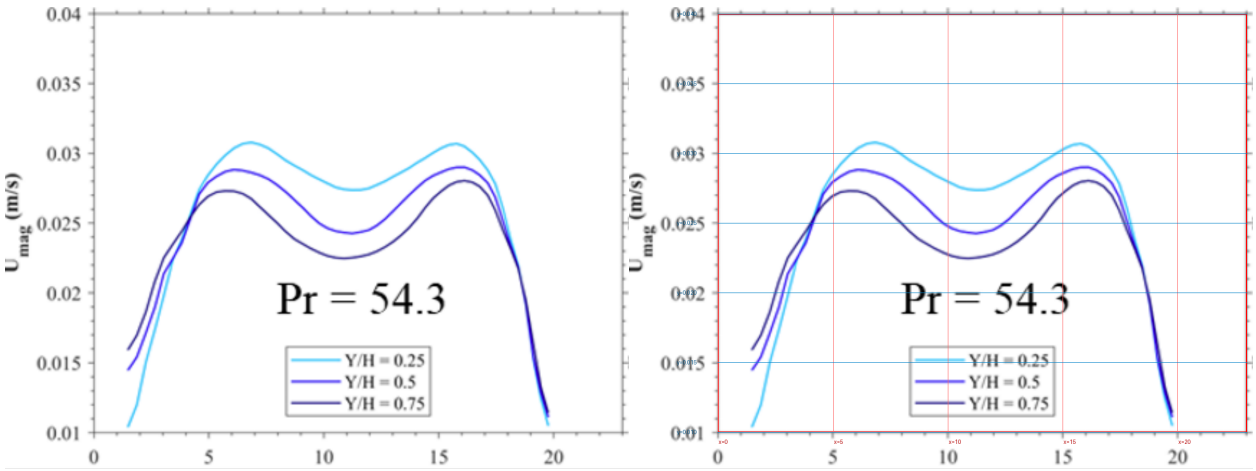


Figure 4.2: Source Image (left) and Calibration Image (right)

Curve Extraction

From there, the source image is used again with the calibrated coordinate frame and the addition of two "masking" frames to allow for the clean extraction of the line data without